

T40g37R15N60 - FASTWIND model diagnostics

Stellar & wind parameters

Stellar

$T_{\text{eff}} = 40,000 \text{ K}$
 $\log g = 3.70$
 $R_* = 15 R_{\odot}$
 $R_{\text{max}} = 120 R_*$
 $T_{\text{min}}/T_{\text{eff}} = 0.6$

Wind

$\dot{M} = 1 \times 10^{-9} M_{\odot} \text{ yr}^{-1}$
 $v_{\infty} = 1500 \text{ km s}^{-1}$
 $\beta = 1$
 $v_{\text{min}} = 0.1 \text{ km s}^{-1}$
 $v_{\text{trans}} = 0.1 a_s$

Atmosphere

$v_{\text{turb, model}} = 15 \text{ km s}^{-1}$
 $\xi_{\text{He}} = 2$
 $f_{\text{cl, max}} = 1$
 $v_{\text{cl, start}}/v_{\infty} = 0.15$
 $v_{\text{cl, max}}/v_{\infty} = 0.3$

Derived parameters

$\log(L/L_{\odot}) = 5.715$

$M_{\text{spec}} = 41.12 M_{\odot}$

$\log Q = -15.528$

Abundances

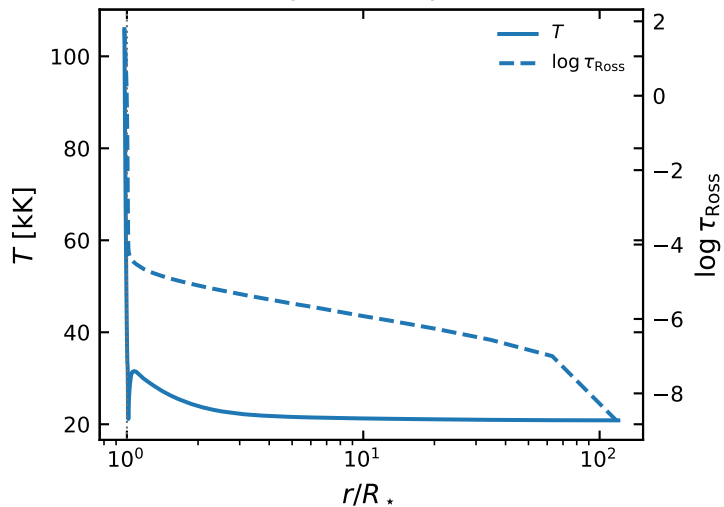
Global

$Z = 0.14 Z_{\odot}$
 $N_{\text{He}} = 0.09$

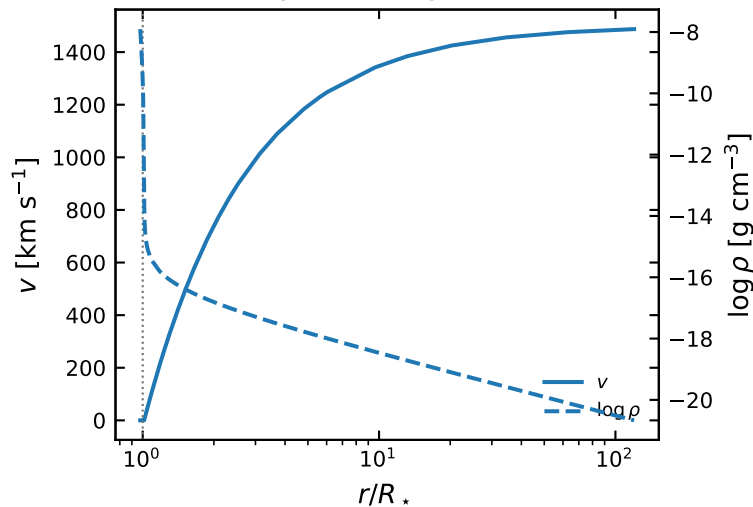
Individual

$\epsilon_{\text{N}} = 6.00$

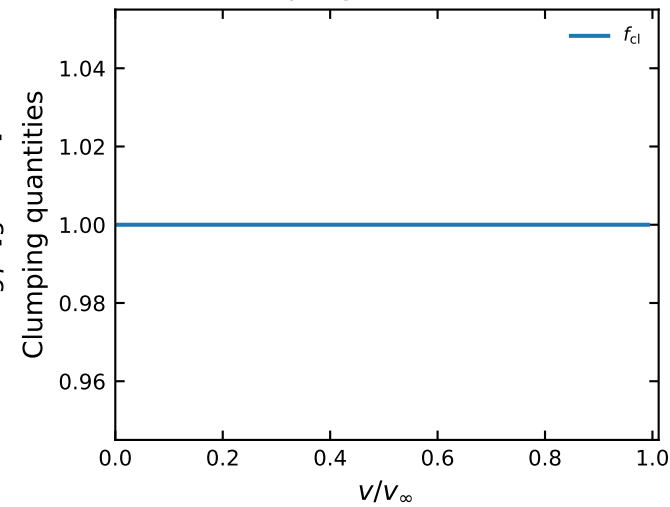
Thermal / optical-depth structure



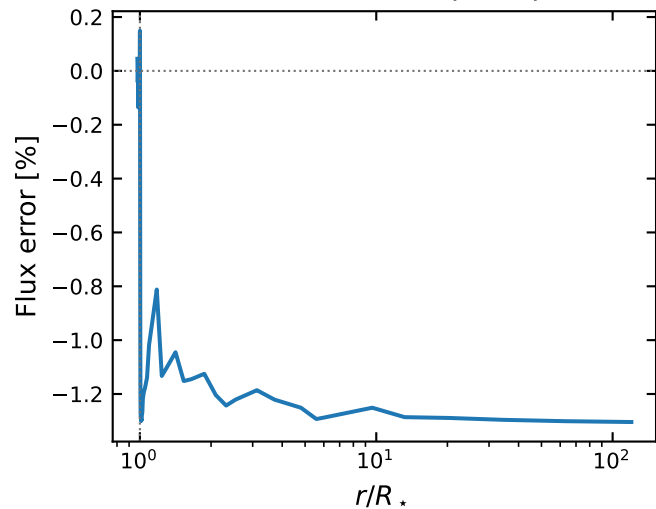
Velocity / density structure



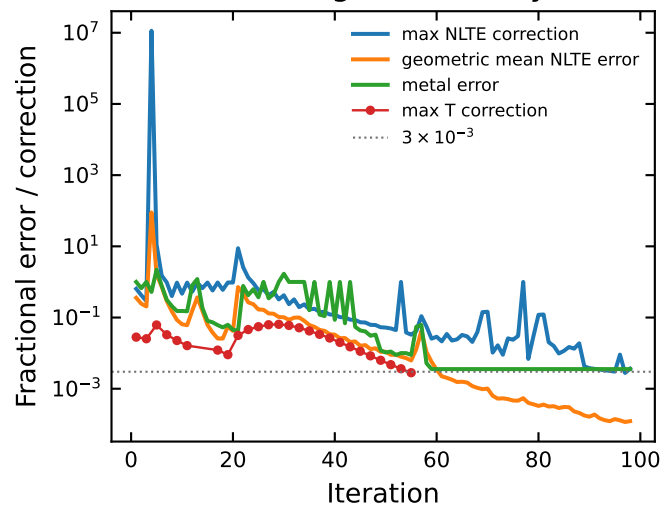
Clumping stratification



Flux conservation (max $|\Delta F/F| = 1.30\%$)



Convergence history



Radiative acceleration

